

HealthConnect Clinic -- Integrated ML Pipeline Testing & Reliability Package

Week 7 -- Machine Learning Engineering Track

Programme	AnalystLab Africa Experience Lab
Week	Week 7
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Track	Machine Learning Engineering
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1. Week 6 → Week 7 Transition

- Main Week 6 output: a swappable NoShowModelInterface with input/output validation, an error analysis on the baseline, a 3-model comparison (LogisticRegression selected over RandomForest/GradientBoosting), and a Slack exchange with Alia Al-Qadri (Data Science).
- Most important Week 6 component: NoShowModelInterface itself -- the mechanism meant to let any model plug into the pipeline without touching ingestion/feature code.
- Main issue/limitation from Week 6: the cross-track handoff was incomplete -- Alia's fitted model had not been received. She has since replied in full with her methodology.
- Component/workstream requiring testing this week: the model-integration interface -- its reproducibility, its input/output validation under harder edge cases, and (newly relevant) whether it can host a model whose feature set differs from ours, since Alia's spec adds one feature (booking_month) ours doesn't have.
- Tracks relevant to Week 7 testing: ML Engineering (own pipeline) and Data Science (Alia -- testing the booking_month delta).
- What was tested: training reproducibility; preprocessing edge cases; invalid/malformed input handling; whether the interface can host a model with a different feature contract; whether booking_month changes model performance.
- What was expected to improve/validate: clearer error handling for malformed input; a generalised interface that doesn't assume every model shares one hardcoded feature set; a real, evidence-based answer on booking_month.

2. Part 1 -- Testing Readiness

A. Component identified

- Component tested: NoShowModelInterface (model-integration layer) and the engineer_features() preprocessing step.
- Intended purpose: let ingestion/feature code stay untouched while different models -- ours or another track's -- plug in, and fail loudly/clearly on bad input.

- Requirement it should satisfy: Week 6 task list items on preprocessing/model interaction and input/output validation.
- Tracks connected: ML Engineering, Data Science.

B. Testing objective

Does the pipeline (a) reproduce identical results on retraining, (b) handle edge-case and invalid input cleanly, and (c) correctly host a model that needs a feature Alia's spec has but ours doesn't?

C. Evidence produced

- Updated Colab notebook with 12 executed tests, a refactored model interface, and a full retrain with an added feature.
- A downloaded Testing & Validation Record (HealthConnect_Week7_Testing_Record.csv), reproduced as Section 6 of this document.
- Slack thread with Alia Al-Qadri (#hc-pod-03), screenshotted, documenting her full methodology.

3. Reproducibility Test

The Week 5/6 baseline (LogisticRegression, fixed RANDOM_SEED, same patient-grouped split) was retrained from scratch and compared against the original run.

- Original baseline: accuracy 0.629, ROC-AUC 0.683.
- Retrained (identical configuration): accuracy 0.629, ROC-AUC 0.683 -- identical.

Result: PASS. The documented setup is genuinely reproducible, not dependent on incidental session state.

4. Preprocessing Component Tests

Three preprocessing behaviours were tested directly against `engineer_features()` and the fitted pipeline, beyond what Week 6 covered:

- First-time patients (`previous_appointments == 0`) correctly get `prior_no_show_rate == 0.0`, never NaN/inf -- the division-by-zero guard holds. PASS.
- Rows with an originally-missing `reminder_channel` are correctly filled with "No Reminder" before reaching the model. PASS.
- An unseen categorical value ("Carrier Pigeon" for `reminder_channel`, never seen in training) does not crash inference -- `OneHotEncoder(handle_unknown="ignore")` absorbs it silently and the pipeline still returns valid probabilities. PASS.

5. Invalid & Unexpected Input Testing

Extends Week 6's three original interface tests (empty input, missing column, valid input) with four new scenarios.

5.1 Edge cases that passed cleanly

- Single-row input: the interface correctly scores the smallest possible valid batch. PASS.
- Extra, unrecognised columns present in the input (e.g. `clinic_notes`, `internal_flag`) are ignored without breaking scoring. PASS.

5.2 A real issue found, fixed, and retested

A batch with "age" set to the text value "forty" instead of a number was scored through the Week 6 interface. It crashed with a raw scikit-learn error ("Cannot use median strategy with non-numeric data") several layers down inside StandardScaler -- not a clear, named error from our own code. FAIL.

This is a genuine reliability gap: the Week 6 interface never validated numeric dtypes before handing data to the fitted pipeline.

Fix: a refactored interface, NoShowModelInterfaceV2, validates every numeric column with `pd.to_numeric(errors="coerce")` before scoring, and raises a clear, named `ValueError` identifying the exact bad column and value.

Retest: the same malformed input now raises "[LogisticRegression (Week 5 baseline)] Column 'age' contains non-numeric value(s): ['forty']" -- a clear, actionable error. PASS.

6. NoShowModelInterfaceV2 -- Refinement Summary

Beyond the dtype fix, NoShowModelInterfaceV2 makes one structural change: `feature_fn`, `feature_cols`, and `numeric_cols` are now constructor parameters instead of hardcoded globals (`engineer_features`, `FEATURE_COLS`, `NUMERIC_FEATURES`). Every existing call site keeps working unchanged (they simply don't pass the new parameters and get the Week 6 defaults), while a model with a different feature contract can now be wrapped correctly.

- Backward compatibility retest: the Week 5/6 LogisticRegression and RandomForest models were re-wrapped in V2 and scored identically to Week 6. PASS.
- Logging was added (`logger.info` on every scoring call) so pipeline activity is now visible in execution logs, not silent.

7. HC-POD Mandatory Cross-Track Testing -- Data Science (Alia Al-Qadri)

Alia's methodology, shared in #hc-pod-03 and confirmed in full on 12 September, matches this pipeline almost exactly: the same missing-value handling (`reminder_channel` → "No Reminder"; `distance_to_clinic_km` and `waiting_time_minutes` → median imputation), the same `GroupShuffleSplit` on `patient_id` (~80/20, `random_state=42`, zero patient overlap confirmed on both sides independently), the same one-hot encoding + numerical scaling, and the same LogisticRegression baseline. The one genuine difference: she engineered `booking_month` from `booking_date`, a feature this pipeline does not use. That is the real, testable dependency this section addresses -- not communication for its own sake, but an actual test with a measurable outcome.

- Track collaborated with: Data Science (Alia Al-Qadri).
- Project dependency: whether Alia's independently-derived feature set changes this pipeline's model, and whether the pipeline's integration point can host a model built on her exact feature set.
- Component/output being tested: the LogisticRegression classifier (with and without `booking_month`) and the NoShowModelInterface/V2 integration layer.
- Information/output received: Alia's full, confirmed methodology -- target definition, missing-value handling, feature engineering (including `booking_month`), excluded predictors, split strategy, and preprocessing choices.
- Information/output provided: the model-agnostic interface design built in Week 6, specifically intended to accept a model matching her approach.

- Testing activity completed: (1) retrained the baseline with booking_month added and compared metrics on the identical held-out set; (2) wrapped a model requiring booking_month in the Week 6 interface to test compatibility, then retested through NoShowModelInterfaceV2.
- Issue/finding identified: booking_month made no meaningful difference to model performance (see table below); separately, the Week 6 interface could not host a model needing this extra feature at all ("columns are missing: {'booking_month'}").
- Refinement/action taken: booking_month is not adopted into the production feature set -- documented as tested-and-rejected, the same honest treatment Week 6 gave the tree-based models. The interface was generalised (NoShowModelInterfaceV2) to remove the underlying compatibility gap.
- Retest result: NoShowModelInterfaceV2 successfully scores the booking_month model (3/3 rows, valid probabilities) where the Week 6 interface failed outright. PASS.
- What changed as a result: the pipeline's model-integration point is no longer tied to one feature set; it can genuinely host whatever feature contract a Data Science model arrives with -- validated against Alia's real methodology, not a hypothetical.
- Evidence: Slack thread in #hc-pod-03 (screenshotted, dated 8 and 12 September); this notebook's Week 7 cross-track testing cells; the Testing & Validation Record (Section 8 below).
- How this improves the overall HealthConnect solution: the ML Engineering pipeline can now absorb a genuine handoff from Data Science on the first attempt, whatever that model's exact feature set turns out to be -- removing a real, previously undetected integration risk before Week 8.

Model comparison -- booking_month feature test

Metric	Baseline (Week 6, no booking_month)	With booking_month (Data Science spec)
Accuracy	0.629	0.622
Precision	0.630	0.621
Recall	0.627	0.625
F1	0.629	0.623
ROC-AUC	0.683	0.682

Conclusion: booking_month does not improve ROC-AUC (0.682 vs. 0.683) and slightly lowers accuracy (0.622 vs. 0.629). This is a genuine, evidence-based negative result, not a failed test -- it is reported honestly rather than assumed to help simply because it came from another track.

8. Testing & Validation Record

All 12 rows below were generated by an automated test harness (record_test()) as each check actually ran -- not written up after the fact.

Testing & Validation Record (continued, landscape)

Component Tested	Testing Objective	Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken	Retest Result
Training step (LogisticRegression baseline)	Confirm reproducibility of the documented setup (fixed RANDOM_SEED, same split).	Retrain the Week 5/6 baseline from the same train/test split and hyperparameters.	Identical metrics to the original run.	accuracy 0.629, roc_auc 0.683 -- identical to Week 6.	PASS	None.	No action required.	N/A
engineer_features() -- prior_no_show_rate	Confirm the division-by-zero guard for first-time patients still holds.	Run on a patient with previous_appointments == 0.	prior_no_show_rate == 0.0, not NaN/inf.	0.0 for all first-time-patient rows tested.	PASS	None.	No action required.	N/A
engineer_features() -- reminder_channel	Confirm missing reminder_channel values are filled, not left null.	Run on rows where reminder_channel was originally missing.	All rows have reminder_channel == "No Reminder".	"No Reminder" in every case.	PASS	None.	No action required.	N/A
Preprocessing pipeline -- OneHotEncoder(handle_unknown="ignore")	Confirm an unseen categorical value doesn't crash inference.	Score a batch where reminder_channel = "Carrier Pigeon" (unseen).	Pipeline handles it silently; valid probabilities returned.	No crash -- probabilities in [0,1] returned (e.g. 0.64, 0.78).	PASS	None.	No action required.	N/A
NoShowModelInterface.predict_risk()	Confirm the interface works on the smallest valid batch size.	Score a single-row input.	Returns exactly 1 row with a valid probability and risk band.	1 row returned, valid probability + risk band.	PASS	None.	No action required.	N/A
NoShowModelInterface.predict_risk()	Confirm unrecognised extra columns don't break scoring.	Score a batch with two extra columns not used by the model.	Extra columns ignored; scoring succeeds normally.	3 rows returned, extra columns ignored.	PASS	None.	No action required.	N/A
NoShowModelInterface.predict_risk() -- Week 6 version	Confirm malformed numeric input fails with a clear, named error.	Score a batch where "age" contains text ("forty") instead of a number.	A clear ValueError naming the interface and the bad column.	ValueError: "Cannot use median strategy with non-numeric data" -- a raw sklearn error, not ours.	FAIL	Week 6 interface has no dtype validation; a bad numeric value is only caught deep inside StandardScaler.	Refactor the interface to validate numeric dtypes before scoring (NoShowModelInterfaceV2).	See retes below.
NoShowModelInterfaceV2 -- backward compatibility	Confirm the refactored interface still works for the original Week 6 models.	Wrap the Week 5/6 LogisticRegression and RandomForest models through V2.	Both score normally, identical behaviour to Week 6.	Both models scored 3/3 rows without error.	PASS	None.	No action required.	N/A

Component Tested	Testing Objective	Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken	Retest Result
NoShowModelInterfaceV2.predict_risk() -- Week 7 fix	Confirm the dtype fix produces a clear, named error.	Re-run: score a batch where "age" contains text ("forty").	A ValueError naming the interface and the 'age' column specifically.	ValueError: "[LogisticRegression (Week 5 baseline)] Column 'age' contains non-numeric value(s): ['forty']".	PASS	None -- this is the retest of the row above.	N/A	PASSED.
Feature set -- booking_month (Data Science spec)	Determine whether Alia's independently-engineered booking_month feature changes performance.	Retrain LogisticRegression with booking_month added, identical split/hyperparameters otherwise.	A measurable, reported change (either direction) in ROC-AUC vs. the Week 6 baseline.	roc_auc 0.682 vs. baseline 0.683; accuracy 0.622 vs. 0.629 -- essentially no change, marginally lower.	PASS	None -- outcome reported either way, not assumed.	Not adopted: no meaningful improvement. Documented as tested-and-rejected, same treatment as Week 6's tree models.	N/A -- first test of this feature.
NoShowModelInterface (Week 6 version) -- cross-track compatibility	Confirm whether the Week 6 interface can host a Data Science model with a different feature set.	Wrap the booking_month model in the Week 6 interface and score a sample.	FAILS -- Week 6 interface hardcodes FEATURE_COLS/engineer_features.	ValueError: columns are missing: {'booking_month'}.	PASS (failure confirmed the gap)	Week 6 interface assumed every model shares the ML Engineering baseline's exact feature set.	Use NoShowModelInterfaceV2 (per-model feature_fn/feature_cols) instead.	See retest below.
NoShowModelInterfaceV2 -- cross-track compatibility (retest)	Confirm the refactored interface can host a Data Science model with a different feature set.	Re-run the previous test through NoShowModelInterfaceV2 with feature_fn=engineer_features_v2.	Scores successfully -- no crash, valid probabilities returned.	3/3 rows returned, valid probabilities.	PASS	None -- retest of the compatibility gap above.	N/A	PASSED -- model-integration point now generalises, not fixed for one case only

9. Configuration & Dependency Review

No new dependencies were required this week. NoShowModelInterfaceV2 uses only pandas and scikit-learn, already declared for Week 5/6. booking_month is derived using pandas' built-in datetime accessor. config.yaml and requirements.txt are unchanged since Week 6 and remain accurate.

10. Assumptions, Limitations, Risks & Dependencies -- Week 7 Update

10.1 Resolved since Week 6

- No dtype validation on numeric columns -- resolved (NoShowModelInterfaceV2 raises a clear, named error).
- Interface could not host a model with a different feature set -- resolved (feature_fn/feature_cols/numeric_cols are now per-model, not hardcoded globals).
- Whether Alia's methodology actually differs from ours in a way that matters -- resolved: confirmed it matches almost exactly, with one isolated, tested difference (booking_month).

10.2 Still open

- Alia's actual fitted model/artifact has still not been received -- she has now confirmed her full methodology but offered a processed dataset and notebook rather than a serialized model. This remains the single largest open item for Week 8.
- No fairness evaluation has been run on the model's use of gender/age features (carried over from Week 6).
- The Sunday-appointment / Knowledge Base inconsistency identified in Week 5 remains unresolved.

10.3 New this week

- booking_month does not improve performance -- confirmed by direct test, not assumption. The current feature set already captures the available linear signal, consistent with Week 6's tree-model finding.

11. Week 8 Readiness Recommendations

- Follow up with Alia specifically for a serialized model artifact (.joblib/.pkl) rather than a methodology description, now that the description itself has been validated as compatible.
- Run full end-to-end validation of the integrated pipeline once a real Data Science artifact is available.
- Incorporate the Data Analytics track's segment-level findings into the feature review, if shared before Week 8.
- Complete a fairness pass on gender/age usage before final integration.
- Update README.md and the repository to reflect NoShowModelInterfaceV2 as the current integration contract.